

MARRAKECH MENARA, Morocco

WMO: 602300

Lat: 31.607N

Lon: 8.036W

Elev: 468

StdP: 95.83

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.0	(c) 5.2	(d) -6.5	(e) 2.3	(f) 22.4	(g) -4.0	(h) 2.9	(i) 20.7	(j) 8.3	(k) 13.9	(l) 6.7	(m) 13.6	(n) 1.3	(o) 80	(p) 0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.5	(c) 42.1	(d) 19.9	(e) 40.0	(f) 20.0	(g) 38.0	(h) 19.9	(i) 22.7	(j) 35.2	(k) 21.9	(l) 34.2	(m) 21.3	(n) 33.0	(o) 4.2	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.8	25.6	18.5	14.1	25.2	17.9	13.6	24.9	68.9	35.5	66.2	34.1	63.8	33.2	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.5	(b) 6.3	(c) 5.4	(e) 1.7	(f) 44.8	(g) 1.8	(h) 1.6	(i) 0.4	(j) 45.9	(k) -0.7	(l) 46.8	(m) -1.7	(n) 47.7	(o) -3.0	(p) 48.8		
			DB													
			WB													
				0.5		24.1	1.8	1.7	-0.8	25.3	-1.8	26.2	-2.8	27.2	-4.1	28.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	12.6	14.2	17.2	19.1	22.4	26.0	29.1	29.3	25.5	22.5	17.1	13.7
	DBStd	6.48	2.41	2.69	3.27	3.40	3.94	3.49	3.42	3.14	2.89	3.11	3.18	2.23
	HDD10.0	9	6	2	0	0	0	0	0	0	0	0	0	2
	HDD18.3	600	177	116	61	29	7	0	0	0	0	4	61	144
	CDD10.0	3937	87	121	223	273	383	481	592	597	464	387	213	116
	CDD18.3	1486	0	1	26	52	132	231	334	339	214	133	24	1
	CDH23.3	17973	14	60	366	629	1491	2751	4489	4463	2224	1230	239	17
(14) Wind	WSAvg	2.4	1.8	2.2	2.6	2.7	2.8	2.9	2.7	2.7	2.5	2.2	1.9	1.7
	PrecAvg	226	26	27	37	25	10	3	2	5	15	20	30	26
	PrecMax	432	125	107	145	118	59	27	32	44	79	69	146	77
	PrecMin	118	0	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	82	29	27	31	30	13	6	6	9	25	18	30	20
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	27.2	32.2	35.2	40.1	42.9	44.8	44.1	40.1	36.0	30.2	25.2
		MCWB	12.6	13.2	15.0	15.9	18.1	19.1	20.3	20.0	19.4	17.9	15.7	13.6
	2%	DB	22.8	25.1	29.2	32.2	37.0	39.6	42.7	41.8	37.1	33.9	28.2	23.1
		MCWB	11.9	12.8	14.6	16.0	18.0	19.2	20.3	20.2	19.4	17.4	14.9	13.0
	5%	DB	21.0	23.2	27.1	29.8	33.9	36.9	40.1	39.3	34.9	31.9	26.2	21.4
		MCWB	11.7	12.8	14.0	15.6	17.6	19.4	20.3	20.6	19.2	17.3	14.7	12.8
	10%	DB	19.1	21.2	25.1	27.0	31.0	34.2	37.8	37.2	32.8	29.8	24.0	19.9
		MCWB	11.6	12.2	13.8	15.4	17.2	19.1	20.3	20.3	19.0	17.2	14.5	12.3
	0.4%	WB	15.2	16.5	17.3	19.0	20.5	22.6	24.5	23.7	22.1	21.1	19.6	16.4
		MCDB	20.0	21.1	24.8	29.2	31.6	35.3	37.4	35.7	32.9	29.0	23.7	19.9
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	14.1	15.0	16.3	17.7	19.5	21.4	22.7	22.7	21.3	19.9	17.5	15.1
		MCDB	19.2	20.6	24.2	27.5	30.9	33.5	36.2	34.8	31.0	27.4	23.3	19.6
	5%	WB	13.1	14.1	15.5	16.7	18.7	20.5	21.8	22.1	20.6	19.1	16.3	14.1
		MCDB	18.2	20.3	23.5	26.0	29.9	32.5	34.7	34.3	29.7	26.6	22.5	18.8
	10%	WB	12.1	13.1	14.7	15.9	17.9	19.8	21.1	21.5	20.0	18.4	15.4	13.2
		MCDB	17.2	19.3	22.4	24.7	28.7	31.4	33.8	33.3	28.9	25.7	21.7	18.1
(35) Mean Daily Temperature Range	MDBR		12.3	12.4	12.8	12.6	13.6	15.1	16.4	15.5	13.2	12.3	11.8	12.2
	5% DB	MCDBR	15.7	16.2	16.6	16.7	17.6	17.6	17.9	17.2	16.1	15.6	14.9	15.0
		MCWBR	7.4	7.0	5.9	5.2	5.2	4.7	4.6	4.3	4.2	4.9	5.8	7.1
	5% WB	MCDBR	12.3	13.2	13.6	14.6	15.7	16.1	16.9	15.8	13.8	12.7	11.5	11.5
		MCWBR	6.6	6.4	5.6	5.2	5.1	4.9	4.9	4.5	4.2	4.5	5.1	6.2
(40) Clear-Sky Solar Irradiance	taub		0.304	0.348	0.413	0.411	0.447	0.488	0.558	0.539	0.453	0.410	0.347	0.312
	taud		2.508	2.361	2.134	2.183	2.089	1.984	1.806	1.871	2.106	2.226	2.421	2.513
	Ebn at Noon		926	908	865	881	848	809	751	760	816	826	864	897
	Edh at Noon		87	110	149	148	163	181	215	199	151	125	94	82
(44) All-Sky Solar Radiation	RadAvg		3.60	4.45	5.62	6.60	7.26	7.87	7.68	6.93	5.93	4.67	3.71	3.28
	RadStd		0.24	0.34	0.29	0.40	0.48	0.36	0.25	0.23	0.30	0.22	0.26	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.46	N/A	-1.38	N/A	N/A	N/A	N/A	+153	+171
(47) Regional (6 neighbors)		+0.32	N/A	-1.19	+0.71	-0.38	N/A	N/A	+153	+171

Nomenclature: See separate page